

COMPUTER FORUM

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MSX and SPECTRAVIDEO

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EDITORIAL

----- SALE -----

Prior to this magazine going out we had a bit of a sale here in W.A., we managed to get hold of 30 Brand New Sony Hit bit computers. We sold them including software for $160.00. So first up I would like to say hello to all our new people that bought them from us.

---- MUSIC COMPETITION ----

The music competition that we ran, attracted 6 entries in all, which we feel is a very poor response, considering we had letters from 2 people to say what about running one, and those 2 did not enter. If you have an idea for a competition please send it in because we will run another one. The results for the Music Competition will be in the next issue.

------ NEWS ------

Just to keep you up to date. At the moment we are in contact with England and Holland with Holland coming up with an idea of putting all Public Domain Software on disk from around the world, so to all you User Groups please send over a disk of all your MSX Public Domain software and I will send it off to Holland. Please send you disk as soon as possible and in return you will receive a disk from Holland. There is also good news for SVI users with the people in holland sending me a list of SVI groups over there. Hopefully there will be more news by the next issue.

------ SOFTWARE SWAPPING ------

I have not figured out how to work out the swapping idea yet, but we do have about 80 titles to swap if anyone can think of a good way of implementing the software swap.

C O V E R W I N N E R

This issues cover winner is -

-- Jared Hyland of New Zealand --

ADVENTURES ANONYMOUS



WITH YOUR GUIDE ALAN CRAWFORD

THE MYSTERY OF LETHE CASTLE.

PREFACE: Every hundred years strange happenings occur, that strikes fear into the very souls of the local villages. Following mysterious noises from the castle during the night, the bodies of three villages are found.

Thats how the story starts, but as in most adventures the story behind the game is of little importance, but serves only to give a certain feel to the player who is about to embark on an adventure of a life time.

I realise that you must all be sitting in your lounge room chairs deeply engrossed in this issue of the magazine and in particular this adventure column, waiting with baited breath for me to tell you when THE MYSTERY OF LETHE CASTLE will be released, so that you can rush off and post your order for the game to the magazine. Well the answer is NOW!!! Yes THE MYSTERY OF LETHE CASTLE is now available exclusively through this magazine at the extraordinary low price of 15 dollars. No that is not a syntax error, just 15 dollars will buy you hours and hours of total enjoyment and pleasure. But this extraordinary low price is for a limited time only. This offer expires on the 15.8.88. When the purchase price must return to 20 dollars. So be quick and order your copy of THE MYSTERY OF LETHE CASTLE now. Remember, in years to come you will be able to tell your children that you where one of the first to play THE MYSTERY OF LETHE CASTLE. What a proud moment in your life that would be. Just imagine sitting next to your 10 year old great great grandson, while he enters machine code programs into his new MSX99, and with a loving smile, turns to you, and says, "please tell me more about THE MYSTERY OF LETHE CASTLE".

I did mention in the last issue that I may be able to organize some sort of competition to coincide with the release of my adventure game, in case you have forgotten its called THE MYSTERY OF LETHE CASTLE, but time was against us and it was thought that by reducing the price for a limited time that everybody who bought the game would be a winner. Depending on sales and of the interest generated by this game I will be continuing with another adventure game that I am writting at the moment. When it is completed anyone who bought the first one will have there name entered into a draw for a free copy of the next game. But this does depend on sales of the first game to see whether it is worth continuing writing the second one. If everything goes ok then the next one will be bigger and better than the first. I have spent a lot of time lately streamlining the original program and cutting out unnecessary routines to give extra space for more puzzles and more locations.

I have just received two adventure games as this magazine went to print they are - JOURNEY TO THE CENTRE OF THE EARTH (graphic adventure) of which I did read a little bit about in the defunct English MSX COMPUTING magazine some time back, but although they thought it was a very easy game I am looking forward to playing it and letting you know the results in an up coming issue of this magazine. I am also looking forward to seeing the other adventure game which is by TONY from LPG. LPG are supplying both of these games to me for review and I hope to hear from them more often. Obviously I can't afford to buy all the adventures that are available so one must rely on the generosity of others to come to the party. As far as adventure games are concerned it looks like LPG are coming to the party (good on them). TONYS adventure is called STRANDED and is a text adventure. I will mention more about them in my next column.

This is the last of the DOTSOFT adventures that I will be looking at as the company as been defunct for some time now and this is the only game from them that I have not completely covered

SKULL ISLAND.

You stand on the beach of SKULL ISLAND a gentle sea breeze ruffling your hair, in your hand is a map. A map left to you by that old sea dog BLACK PATCH JONES. Ahead jungle stretches to the base of a steep cliff rising to a high PLATEAU. Glancing at the map you see cryptic signs, palms a cross and a chest. Strange how some people also call this place TREASURE ISLAND. Whats this.... A parrot with a pegleg just flew past.... as the adventure begins.

Where you start there is a rowboat (same one that was in the MAD MUMMY), sand and a cross. You will also find an ANCHOR if you 'LOOK ROWBOAT'. South takes you to the steep cliff where you find a SHELL and a TREE. Take the shell with you NORTH and then WEST to the end of the beach by a SIGN and some ROCKS. Do exactly what the sign says and then DIG (you need the shell for this). You will find a BOTTLE and a ROPE, in the BOTTLE there is a NOTE, 'GET' then 'READ' the NOTE. From here SOUTH takes you to another steep cliff where you will find JUNGLE and a WATERFALL. Of which neither have any use. East, and you are back at the TREE, but first you must go EAST again to yet another steep cliff where you should 'TIE ROPE' to 'ANCHOR' and then throw the ROPE which catches on the LEDGE. 'CLIMB ROPE' to the LEDGE to find an AXE and some FRUIT (the FRUIT is needed later on). 'GET AXE' and 'CLIMB ROPE' again, then go west to the TREE. The TREE is your way to the next part of the game. 'CHOP TREE' four or five times to fell the TREE and cut rungs in it. Then 'CLIMB TREE'.

You are now beside a wide lagoon. A cave can be seen across the other side of the lagoon but you will need an old cabin DOOR to get across. East from here takes you to a rocky bay where you find ROCKS, SHIPWRECK and some DEBRIS, amongst the DEBRIS you will find the DOOR. From here 'GO ROCKS' to the SHIPWRECK, on board the deck of the wreck is a mast (the program makes no mention of this). 'CLIMB MAST' to the crows nest to find a PATCH and a CORKSCREW. Then back down and EAST to the cabin where you will find a SKELETON and a HAT. 'LOOK SKELETON' to find a COAT and a LOCKET. East again to the HOLD and a LAMP and COIL (the COIL is a fuse). You will need to make two trips to the SHIPWRECK to collect everything.

Back to the wide lagoon and then WEST to a DAM. (I'll get back to the DAM shortly). WEST to a path and WEST again take you to a STOCKADE WALL, where you must 'CHOP STOCKADE' four times to open up a hollow. The CHAIN inside the hollow should be pulled to open up the entrance to the STOCKADE. SOUTH take you inside stockade where you find a KEG with a PLUG. 'REMOVE PLUG' (CORKSCREW NEEDED) and FILL LAMP. Then SOUTH, go inside a rough hut. You will need the THONGS and you must 'MOVE ASHES' that you find in the HEARTH. In the ASHES there is a CASKET which can be opened with the LOCKET. Inside the CASKET are some CRYSTALS. Return to the LAGOON.

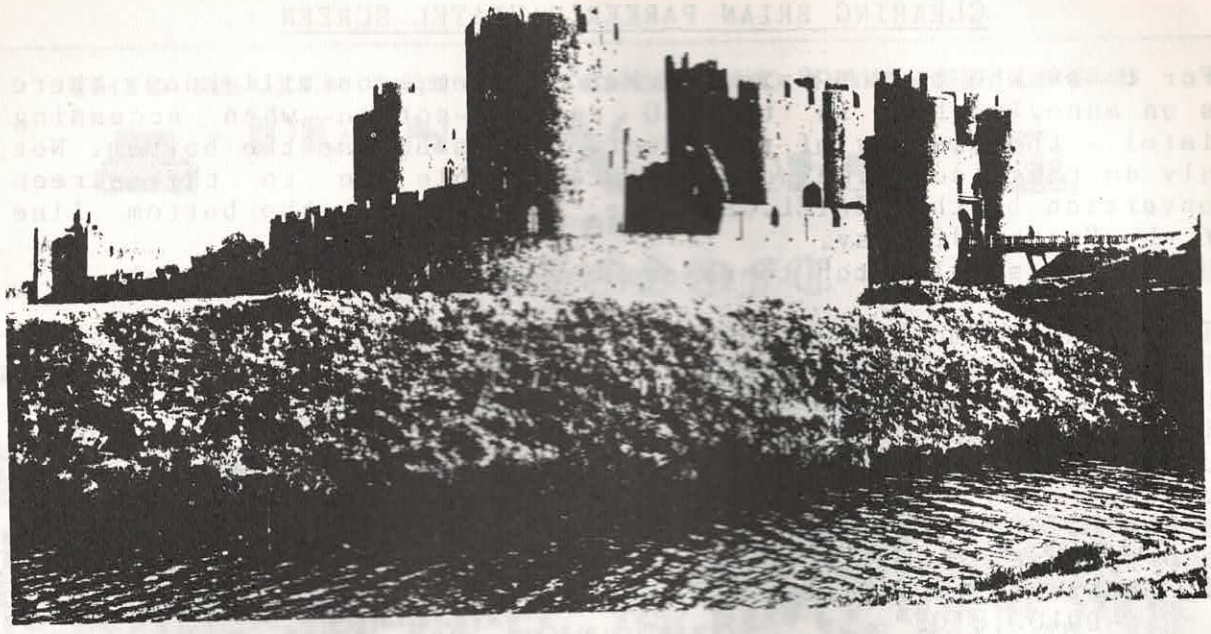
By the lagoon you should 'DROP DOOR' and 'GO DOOR', you will need to take the lighted LAMP, AXE, THONGS and all the clothes with you. Once in the cave drop the LAMP and leave it there. Then EAST to a sandy path, SOUTH to a rocky harbour and SOUTH again to a bamboo thicket. Here you find BAMBOO which you 'CHOP' and then 'TIE' with the THONGS to make a LADDER, this needs to be dropped in the cave for your escape. Before going any further you must 'WEAR' all the clothes that you have found in various places throughout the game. Then off we go SOUTH to the pirates camp. Failure to wear all the clothes results in sudden death.

In the middle of the pirates camp you will find GUNPOWDER and BOXES (ignore the BOXES), 'GET GUNPOWDER' and return to the dam where you will 'FIT COIL', 'LIGHT COIL', (with CRYSTALS) and exit that location speedily. When you return the dam has been blown up to reveal a CREVISE in which you will find a key. Then return to the pirates camp and go WEST to a narrow path by a HILL and OAK TREE. West from here is a clearing full of CANNIBALS (dont go here). Instead 'CLIMB OAK' and 'SWING VINE' which you find there. This takes you to a jungle path, NORTH of which you find SKULL ROCK.

At SKULL ROCK you must DANCE A JIG to make the skull laugh and enable you to see a PLATE which has a KEYHOLE. While you are doing all this you will hear screeching monkies, you will need to drop the FRUIT to get rid of them. Once you find SKULL ROCK you dont have too many moves up you sleeve before the cannibals come, so dont waste to many moves looking around. 'UNLOCK PLATE', you will then find the CHEST 'OPEN CHEST' for the GOLD. After you have the GOLD, 'MOVE CHEST' to reveal a HOLE, and 'GO HOLE'. If you dropped the ladder in the cave then you are home free, if not you will be the main course at dinner tonight. Take the TREASURE back to the beach and drop it where you saw the CROSS, this then completes SKULL ISLAND, easy wasn't it.

Well thats all for this time but before I go I would like to thank BERT ATTWOOD from my local user group for loaning me a couple of adventure games that I had not played. Dont forget to drop me a line if you are having any problems in an adventure I will try to be of assistance. Dont forget to order your copy of THE MYSTERY OF LETHE CASTLE.

Till next time, bye for now ALAN.



THE MYSTERY OF LETHE CASTLE is a new adventure game by ALAN CRAWFORD.

The game revolves around gaining entry to the castle, solving the problems therein and reaching your end objective. To do this you will need to visit approx 20 locations, but don't be mislead if you think this is to few, as you will need to retrace your steps on more than one occasion.

Although graphics to me are not terribly important LETHE CASTLE is as good as any other MSX adventure I have played.

The problems, which there is an ample supply of, were very good indeed. The solutions were logical, without being too easy or too complex and showed a great deal of originality.

Considering this is ALAN'S first adventure game, this player was very impressed indeed and eagerly awaits the next one.

Reveiw supplied by - PETER ROBINSON.

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CLEARING BRIAN PARKER'S VIATEL SCREEN

For those who bought Brian Parker's modem, you will know there is an annoying bug in the 40 column screen when accessing Viatel - the display of the function keys across the bottom. Not only do they look like nothing intelligible due to the screen conversion by the VIATEL.CHR file, they take up the bottom line of the Viatel display.
Here is a small patch to remove the function keys entirely.

There is a memory location (FA06) used for protecting screen lines from the bottom up. The algorithm used is 256-n, and as the location normally has &HFF in it, only 256-255 (the bottom line) is protected. Changing this to 0 'unprotects' the whole screen, allowing a screen clear command to clear **ALL** of the screen.
Using DDT80.COM (or DDT.COM), have a look at the first 16 memory locations of the VIATEL.COM programme.
It should be as follows:

```
-D0100,010F
0100 2A 06 00 F9 00 00 C3 0C 01 C3 00 00 CD 62 03 CD
```

&H0106-7-8 is a JUMP to &H010C, which is only avoiding a jump to a warm boot. The two address bytes (0C 01) will be changed shortly. for the moment, look at &H20E0:

```
-D20E0,20FF
20E0 00 00 C1 FE FF C0 3E 0B 32 3F 06 2B C9 1A 1A 1A
20F0 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A 1A
```

Note the string of '1A's from 20ED on. This is where the patch will go. Enter the following:

```
-S20ED
20ED 1A 21
20EE 1A 06
20EF 1A FA
20F0 1A 36
20F1 1A 00
20F2 1A C3
20F3 1A 0C
20F4 1A 01
20F5 1A .
```

This small programme consists of the following:

```
20ED 2106FA   LD HL, FA06   ; put 'bottom up protect' location
                            ; into HL register pair
20F0 3600     LD (HL), 00   ; put 00 into 'bottom up protect'
20F2 C30C01   JP 010C       ; jump to where originally going
```

Now the original jump has to be changed to the patch:

```
-S107
0107 0C ED
0108 01 20
0109 C3 .
```

Enter ^C and at the prompt, type: SAVE 32 (drive):VIATEL.COM

When the Viatel programme is now run, there is no bottom line protection, so those annoying 'function' keys are removed and the full screen can now be used by the incoming Viatel data.

-ARGUSY-

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INPUT

THIS ARTICLE WAS MEANT TO APPEAR IN THE LAST ISSUE, BUT DUE TO A SMALL "ERROR" IT WAS LEFT OUT.
WE HUMBLY APOLIGIZE FOR ANY INCONVENIENCE THAT MAY HAVE BEEN CAUSED BY THIS SMALL OMISSION ON OUR PART.

ONE OF THE MOST COMMON COMPLAINTS THAT WE RECIEVE IS:
'I TYPED IN THE PROGRAM FROM THE MAGAZINE BUT IT WON'T RUN.'
TO TRY AND DE-BUG THE PROGRAM ON THE PHONE IS ALMOST IMPOSSIBLE. IF YOU ENTER IN A PROGRAM AND IT WON'T RUN, THEN *PLEASE* DON'T CALL US AS WE ARE UNABLE TO HELP YOU OVER THE PHONE. THE FIRST THING TO DO IS TO DROP OUT A HARD COPY OF YOUR LISTING TO COMPARE IT WITH THE LISTING IN THE MAGAZINE. MOST OF THE TIME YOU WILL FIND A SIMPLE TYPING ERROR HAS CAUSED THE PROBLEM.

IF YOU ARE UNABLE TO FIND THE ERROR, YOU CAN SEND A TAPE OR DISK TO US AND, FOR A SMALL FEE OF $1.50 PER LISTING, WE WILL SEND YOU A COPY OF THE OFFENDING PROGRAM.

ANOTHER WAY TO HELP OVER COME THIS PROBLEM WAS FOUND BY THE OLD USER GROUP IN TASMANIA, *SAUG*. THEY DEVELOPED A PROGRAM TO READ "ASCII" SAVED FILES, WORKED OUT A CHECKSUM FOR EACH LINE THEN SENT THE CHECKSUM, PLUS THE LINE TO THE PRINTER. THEY THEN WORKED OUT ANOTHER PROGRAM TO ENTER IN A LINE, WORK OUT THE CHECKSUM AND COMPARE IT TO THE CHECKSUM AT THE BEGINNING OF THE LINE. IF ALL IS *OK*, THEN THE LINE WILL BE SAVED TO AN *ASCII* FILE ON EITHER DISK OR TAPE. IF THERE IS AN *ERROR* IN THE LINE THEN THE LINE WILL *NOT* BE SAVED AND YOU WILL HAVE TO FIND THE FAULT BEFORE THE PROGRAM WILL LET YOU SAVE THAT LINE.
'BUT WHAT IF I WANT TO CHANGE SOMETHING IN THE LINE', I HEAR YOU SAY.
ALL YOU HAVE TO DO IS TO ENTER IN A *'SPACE'* INPLACE OF THE *CHECKSUM* CHARACTERS.

__*WARNING*__
THIS WILL BYPASS THE *CHECKSUM* ROUTINE. ENTER IN YOUR LINE *CAREFULLY*.

THE *CHECKSUM* ROUTINE WILL NOT PICK UP SPACES IN THE LINE.
EG.
IT WILL SEE THE LINE:

```
10 PRINT "A BIG DOG"
```

AND THE LINE:

```
10 PRINT"A  BIGDOG"
```

AS THE SAME.
SO TAKE CARE WITH YOUR SPACING.
IT WILL PICKUP:

```
10 PRNIT"A BIG DOG"
```

AND:

```
10 PRINT"A Big Dog"
```

AS ERRORS, SO CHECK TO SEE THAT YOU ENTER IN UPPER *AND* LOWER CASE IN THE CORRECT PLACE, AS WELL AS YOUR SPELLING.

ROSS MOORE SENT IN A COPY OF THE PROGRAMS, UNFORTUNATELY THEY WERE FOR THE *SVI* COMPUTERS ONLY, SO I HAD TO AMEND THEM FOR THE *MSX* RANGE.

AFTER *VERSION 12* OF THE *CHECKSUM* PROGRAM, I HAD IT AT WHAT I THOUGHT WAS *IT*, BUT, OH NO, IN COMES *ROSS* WITH ANOTHER CHANGE FOR THE PROGRAM.

"WHY DON'T WE MAKE IT SO THAT WE CAN SET IT UP TO PRINT OUT THE LISTINGS IN COLUMNS."

(OF COURSE THE *'WE'* MEANS ME) SO *VERSION 16* IS NOW THE FINAL DRAFT.
(*I HOPE!*)

INPUT

THE *INPUT* PROGRAM WAS MUCH EASIER TO CONVERT. *INPUT.BAS* ONLY WENT TO *VERSION 5* BEFORE IT WAS ACCEPTABLE.

EVERYTHING THAT YOU NEED TO KNOW ABOUT THE *INPUT* PROGRAM IS IN THE PRINT STATEMENTS AT THE START OF THE LISTING, AND CAN BE DROPPED OUT TO A PRINTER FOR A HARD COPY BY CHANGING THE *PRINT* STATEMENTS TO *LPRINT* FOR ONE RUN.

FOR THOSE WHO WISH TO ENTER IN THE LISTINGS AS BEFORE, SIMPLY LEAVE THE *CHECKSUM* OUT.

IM OTHER WORDS, IF THE LINE LOOKS LIKE THIS:

AQ 10 REM TITLE SCREEN

ALL YOU NEED TO DO IS ENTER IN:

10 REM TITLE SCREEN

AS YOU WOULD NORMALLY. YOU DO NOT NEED TO USE THE *INPUT* PROGRAM!

UNFORTUNATELY, THIS WILL NOT HELP US WITH THE PROBLEM OF LISTING OUT THE *SVI* PROGRAMS THAT USE THE GRAPHICS CHARACTERS AS WE ARE USING AN *MSX* PRINTER TO DROP OUT OUR LISTINGS. IF YOU KNOW OF AN *SVI* PRINTER FOR SALE, THEN PLEASE LET ME KNOW.

IF YOU HAVE ANY TROUBLE ENTERING IN THE *INPUT* LISTING (*HEAVEN FORBID*) THEN SEND US A BLANK DISK OR TAPE PLUS $1.50 AND WE WILL SEND YOU A WORKING COPY OF THE PROGRAM.

IF YOU COME UP WITH ANY *IMPROVEMENTS*, THEN SEND THEM IN. THE LISTINGS IN THIS MAGAZINE ARE GOOD, BUT THEY *ALL* CAN BE IMPROVED. SO DON'T BE SHY. IF YOU CAN IMPROVE A LISTING, THEN LET US ALL KNOW OF YOUR WORK.

SEND IN YOUR COMMENTS ON THE INPUT PROGRAM.

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MSX BASIC TUTORIAL

PART 5

QUEST

Here is a good little Question and answer program for your children or your freinds.

```
AL  10 CLS:KEY OFF:WIDTH 40:CLEAR 5
    000
AO  20 DIM Q$(50),A$(50)
OM  30 CLS:L$="=":FOR X=1 TO 40:LOC
    ATE X,1:PRINTL$:NEXT X
AM  40 PRINT TAB(6);"1.. ENTER QUES
    TIONS/ANSWERS"
AL  50 PRINT:PRINTTAB(6);"2.. LOAD
    QUESTIONS/ANSWERS"
DE  60 PRINT:PRINTTAB(6);"3.. SAVE
    QUESTIONS/ANSWERS"
AB  70 PRINT:PRINTTAB(6);"4.. CHANG
    E QUESTIONS/ANSWERS"
AE  80 PRINT:PRINTTAB(6);"5.. PLAY
    GAME"
BL  90 PRINT:PRINTTAB(6);"6.. END
AP  100 FOR X=1 TO 40:LOCATE X,17:P
    RINTL$:NEXT X
DN  110 LOCATE 11,15:INPUT"Choose:"
    ;CH
BE  120 P=0
DJ  130 IF CH=6 THEN KEY ON:END
BA  140 IF CH<1 OR CH>6 THEN 100
CE  150 ON CH GOTO 180,360,580,840,
    1070
AE  160 '
CK  170 '''''''''''''''''''''''''''
    '''''''

EJ  180 '' ENTERING OF QUESTIONS /
    ANSWERS
CI  190 '''''''''''''''''''''''''''
    '''''''

BI  200 CLS
DJ  210 PRINT"ENTER A PASSWORD, SO
    ONLY YOU CAN "
AB  220 PRINT"CHANGE THE QUESTIONS
    OR ANSWERS"
EK  230 PRINT:PRINT:INPUT "Password
    :";PW$:CLS
BG  240 X=NQ
EM  250 X=X+1
DI  260 PRINT"QUESTION";X
AM  270 PRINT:INPUT Q$(X)
FB  280 IF Q$(X)="" THEN NQ=X-1:GOT
    O 320
DG  290 PRINT:PRINT"ANSWER";X
BE  300 PRINT:INPUT A$(X)
BO  310 PRINT:PRINT:GOTO 250
BA  320 PRINT"##### DONT FORGET TO
    SAVE THIS #####"
BG  330 FOR T=1TO1000:NEXT T:GOTO 3
    0
AI  340 '
CO  350 '''''''''''''''''''''''''''
    '''''''

CA  360 '' LOAD QUESTIONS / ANSWERS
CM  370 '''''''''''''''''''''''''''
    '''''''

BP  380 CLS
EJ  390 PRINT"   ### LOADING QUESTI
    ON/ANSWERS ###":PRINT:PRINT
CO  400 PRINTTAB(9);"1.. TAPE"
DG  410 PRINT:PRINTTAB(9);"2.. DISK
    "
DI  420 PRINT:PRINTTAB(9);"3.. MENU
    "
CO  430 PRINT:PRINTTAB(11);:INPUT"C
    hoose:";TD
DE  440 IF TD<1 OR TD>3 THEN 380
BN  450 ON TD GOTO 460,490,30
DA  460 PRINT:PRINT:PRINT"PUT TAPE
    INTO RECORDER"
CA  470 PRINT"AND POSITION IT AT QU
    ESTION & ANSWER"
AI  480 PRINT"SECTION":MOTOR ON
DN  490 OPEN "QADATA"FOR INPUT AS 1
AD  500 INPUT #1,NQ,PW$
DF  510 FOR X=1 TO NQ
AJ  520 INPUT#1,Q$(X),A$(X)
DD  530 NEXT X
BF  540 CLOSE #1
DA  550 GOTO 30
AI  560 '
CO  570 '''''''''''''''''''''''''''
    '''''''

EJ  580 '' SAVE QUESTIONS / ANSWERS
CM  590 '''''''''''''''''''''''''''
    '''''''

BE  600 CLS
DG  610 IF NQ=0 THEN LOCATE 6,10:PR
    INT"THERE IS NOTHING TO SAVE":F
    ORT=1TO500:NEXTT:GOTO 30
JE  620 PRINT"   ### SAVE QUESTIONS
    /ANSWERS ###":PRINT:PRINT:PRINT

CN  630 PRINTTAB(9);"1.. TAPE"
DF  640 PRINT:PRINTTAB(9);"2.. DISK
    "
DH  650 PRINT:PRINTTAB(9);"3.. MENU
    "
CP  660 PRINT:PRINTTAB(11);:INPUT"C
    hoose:";TD
EA  670 IF TD<1 OR TD>3 THEN 600
BF  680 ON TD GOTO 690,740,30
CH  690 MOTOR OFF
AN  700 PRINT:PRINT:PRINT"PUT A BLA
    NK TAPE INTO YOUR CASSETTE"
FE  710 PRINT"PLAYER, THEN PRESS PL
    AY AND RECORD"
BH  720 PRINT"THEN PRESS [ENTER] or
```

```
    [RETURN]"
BG  730 INPUT ZZ
FH  740 PRINT:PRINT:PRINT:PRINT:PRI
    NT"          SAVING Q&A DATA":P
    RINT:PRINT"          PLEASE WAI
    T"
DH  750 OPEN "QADATA"FOR OUTPUT AS
    1
AK  760 PRINT#1,NQ,PW$
DB  770 FOR X=1 TO NQ
CC  780 PRINT#1,Q$(X);",";A$(X)
DH  790 NEXT X
AA  800 CLOSE
DH  810 GOTO 30
AP  820 '
DF  830 '''''''''''''''''''''''''''
    '''''''
AP  840 '' CHANGE QUESTIONS / ANSWE
    RS
DD  850 '''''''''''''''''''''''''''
    '''''''
BI  860 CLS
CF  870 INPUT "What is the password
    :";EP$
DK  880 IF EP$=PW$ THEN 900 ELSE 89
    0
AG  890 PRINT"SORRY YOU GOT IT WRON
    G":FORT=1TO1000:NEXTT:GOTO 30
BC  900 '
AO  910 PRINT:PRINT:INPUT"WHAT NUMB
    ER DO YOU WANT TO CHANGE:";WN
AG  920 PRINT:PRINT Q$(WN):PRINT A$
    (WN)
DF  930 PRINT:PRINT:INPUT "CORRECT
    ONE TO CHANGE Y or N";YN$
FE  940 IF YN$="Y" THEN YN$="":GOTO
     960
AB  950 GOTO 910
EA  960 PRINT"ENTER NEW QUESTION AN
    D ANSWER OR JUST  PRESS ENTER T
    O LEAVE UNCHANGED"
EB  970 PRINT:PRINT:PRINT
BM  980 PRINT"QUESTION";WN
AJ  990 INPUT Q$(WN)
AC  1000 PRINT"ANSWER  ";WN
HD  1020 PRINT:PRINT:INPUT"ANY MORE
     CHANGES Y or N";YN$
FH  1030 IF YN$="Y" THEN 910
AF  1040 GOTO 30
CN  1050 '
AH  1060 '''''''''''''''''''''''''''
    '''''''
DO  1070 '' PLAY GAME AND CHECK IF
    CORRECT
AJ  1080 '''''''''''''''''''''''''''
    '''''''
FE  1090 CLS

GJ  1100 P=P+1
BJ  1110 IF P>NQ THEN 30
FP  1120 LOCATE 12,19:PRINT"ENTER
    QUIT' TO END"
JP  1125 LOCATE 1,1:PRINT"TOTAL QUE
    STIONS:";NQ
DC  1130 LOCATE 22,1:PRINT"QUESTION
    :";
FH  1140 TQ=TQ+1:PRINT TQ
JJ  1150 LOCATE 2,4:PRINT"RIGHT:";R
DJ  1160 LOCATE 25,4:PRINT"WRONG:";
    W
EM  1170 PRINT:PRINT:PRINT Q$(P)
EJ  1180 PRINT:INPUT"YOUR ANSWER:";
    YA$
CE  1190 IF YA$="QUIT" THEN TQ=TQ-1
    :GOTO30
MJ  1200 IF YA$=A$(P) THEN R=R+1:FO
    RX=1TO10:BEEP:NEXTX:PRINT:PRINT
    "CORRECT":FORT=1TO500:NEXTT:IF
    Q$(P+1)=""THEN 30 ELSE GOTO 109
    0
BM  1210 W=W+1:PLAY"O2AA":LOCATE25,
    10:PRINT"WRONG":FORT=1TO500:NEX
    TT
BM  1220 LOCATE 0,12:PRINT"CORRECT
    ANSWER IS:";A$(P)
BA  1230 FOR T=1TO1500:NEXT T:GOTO
    1090

   THE END
```

I have received a number of phone calls for various questions, most of which I have answered in my Basic Tutorial, so please make sure I have not allready covered the subject before giving me a ring, and one more thing, to those people in Victoria or in other states remember the time difference between you and W.A. because I'm not used to answering questions at 5 oclock in the morning. Well thats enough wingeing from me onto my tutorial.

The QUEST program was made to show you the use of STRINGS ($)

Again there are to many lines to break down the whole program but I will tell you about the functions you should know about.

LINE 20

```
DIM Q$(50),A$(50)
```

I DIMmensioned our box to hold 50 questions and answers. You can do as many as you like.

LINE 30

```
L$="="
```

This line draws an = sign across the screen.

LINE 40 - 90

Prints up main Menu.

LINE 150

```
ON CH THEN GOTO
```

This line looks at the value of CH .
In other words if CH=1 then goto position 1(180) of if CH=5 then goto position 5(1070)

LINE 280

```
IF Q$(X)="" THEN NQ=X-1
```

To put this line another way -
IF Question (1) or 2 or 3 etc = Nothing then let NQ (Number of Questions)= X (Times through the loop) - 1. Take 1 away from the value of X at the time

LINE 300

```
INPUT A$(X)
```

This line asks for an answer to the question you just typed in.

LINE 480

```
MOTOR ON
```

This is used only for cassette players to make sure you can wind the tape to the required place.
MOTOR OFF turns the cassette player's motor off.

LINE 490

```
OPEN "QADATA" FOR INPUT AS 1
```

This line opens up a file called 'QADATA' and makes it ready to send input into the computer.
1.. INPUT means put-in data to computer.
2.. OUTPUT means put-out data from computer.

LINE 500

```
INPUT #1,NQ,PW$
```

This line puts at the start of the file these 2 variables NQ (No of questions) & PW$ (password)

LINE 510
FOR X=1 TO NQ (No of questions)
Start of counting loop.

LINE 520
INPUT #1,Q$(X),A$(X)
This line loads into memory whatever is stored in Q$(X)(Question NoX),A$(X)(Answer NoX)
Whatever number X is at the time.

LINE 530
NEXT X
Meaning X=X+1 go back to line 510

LINE 540
CLOSE
You must allways CLOSE the file when you have finished reading it or writing to it.

LINE 610
IF NQ=0 THEN ----
If the No of questions = 0 then print up the warning message. This line is put in as a trap just in case you should accidently save nothing on top of your existing file. In other words if you use option 3 when you have your Q & A file complete and you thought you were loading in your file you would really be wiping it out.

LINE 750 - 800
Does the same as loading the file except this option saves it to tape.

LINE 870 - 890
INPUT "Password";EP$
These lines check to se if your password that you typed in under EP$ is the same as PW$ which is stored on tape, if not it tells you.

LINE 1100 - 1160
Prints up the screen to tell you the number of questions you have, Number right and number wrong

LINE 1170
PRINT Q$(P)
Prints up question (P)

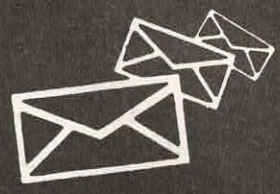
LINE 1180
INPUT "Your answer:";YA$
Stores Your Answer under YA$

LINE 1200 - 1230
Checks to see if your answer (YA$) = the correct answer A$(P) if so it adjusts the scores and then goes back for another question.

Send in your queries or if you are having problems with one of your own programs put it on tape and send it to us with enough for return mail, and we will try to solve you problem.

Bye till next time...

MAILBOX



Here is some help replys from our last issue, sent in by Tony James of Victoria.

To Jeffrey Dean asking about the book "THE MAGIC OF SPECTRAVIDEO try writing to -

INTERSOFT PTY. LTD.
PO BOX 5078
JOHANNESBURG 2000
SOUTH AFRICA

It is an excellant book, especially for anyone contemplating any work with machine code.

Phillip Christie asked about BIOS Calls and Hooks. The above book has a list of hooks which may be of help.

David Stewart asked about double line feeds for an Epson printer I have an Epson LX-80. I can't get it to do double space other than with the Dip switch, but I can set the distance of a line feed by -

```
1PRINT CHR$(27)+"3"+CHR$(?)
```

where ? is the distance of the line feed in inches divided by 180. i.e. CHR$(90) will give you a half inch between lines.

I hope this will help out the people concerned. Why not publish people's addresses when they put articles in the magazine? Maybe some people would like to know if the person asking the questions was living just down the road from them.

REPLY

THAT DECISION IS UP TO PEOPLE THAT WRITE IN, IF YOU WANT YOUR ADDRESS PRINTED IN THE MAGAZINE PLEASE SAY SO IN YOUR LETTER, OTHERWISE IT WILL NOT BE PRINTED.

Here is a few queries we received from D.C. Morgan of South Australia.

If by a stroke of good luck I can purchase a Sony disk drive are there :-

1. Sufficent Disk programs available.
2. Can I use MS/DOS programs which are not MSX.
3. Can I use CPM if I can purchase a Sony disk drive.
4. Being a novice, can I put the programs I have done onto a disk.
5. Is it possible to modify another disk drive to use with my Sony MSX if I cant get a Sony disk drive.

OUR ANSWER

Yes, No, No, Yes, Yes

Let me elaborate a little more on my answers.

1. Yes there are a lot of Disk programs around, if you have not already done so I suggest you write to MSX International in South Australia.
2. As far as running MS/DOS from other machines, it cant be done as far as I know, if I am wrong could someone let me know.
3. Unfortunately the only MSX1 machine to run CPM is the Spectravideo Express 738.
4. You can save as many of your home made programs onto disk as you like.
5. It is possible to modify another disk drive to run on an MSX machine if you have the technical no-how. Please don't ask me how, I draw the line at inserting batteries in a calculator.

Perhaps a reader might want to write in with such a modification

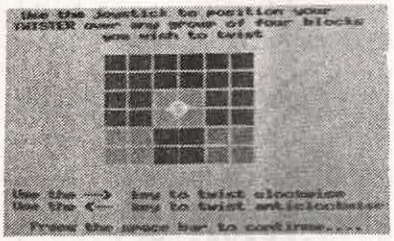
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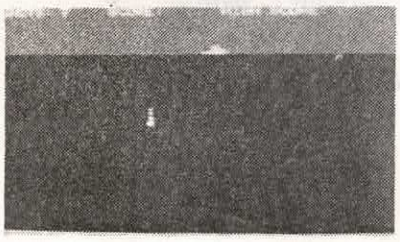
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SVPAK23 - MSXPAK23 -

TWISTER - Like a two dimensional Rubiks cube, this game displays nine different coloured squares each containing four smaller squares. These 36 squares are then jumbled around on the screen and you must rebuild the original display. Its very simple - VERY addictive!

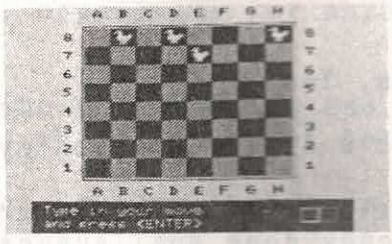
REVERSE - Your task is to arrange the ten jumbled numbers into ascending order by reversing some or all of them. Check it out - its not easy!



TS852 - MSX852 -

GRAND PRIX - You are in the drivers seat in a race to the chequered flag while avoiding slower cars. Hitting another car or two wheels off the track will slow you down and lose points.

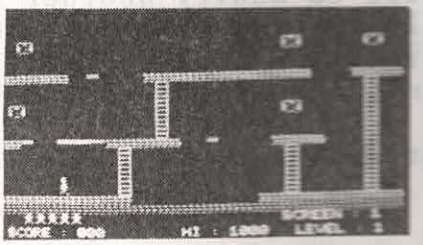
SUBSTRIKE - Drop depth charges on the enemy below. The deeper he is or the faster he moves, the more you will score.



SVPAK24 - MSXPAK24 -

FOX & GEESE - Can you as the Fox penetrate the line of advancing geese and get to the other side of the board before they corner you.

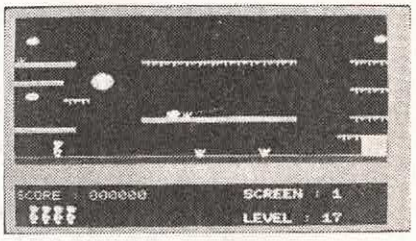
THINK - This is a form of word mastermind. You must guess the word that the computer has selected from the vast number stored in its data banks. This game will certainly make you do as the name suggests.



SVPAK7 - MSXPAK7 -

BOMB SCARE - Very addictive game in which you must climb to different platforms and defuse 5 bombs before your bonus runs out. Has 6 separate screens with the hazards of moving platforms and conveyor belts with holes in them.

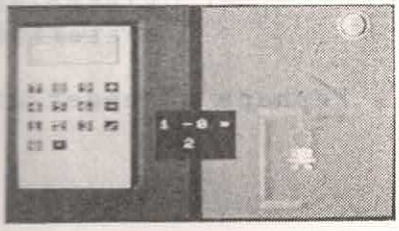
DUNGEON ADVENTURE - A text adventure where you explore the dungeon fighting monsters and gain treasures to discover the location of the great Orb.



SVPAK21 - MSXPAK21 -

GOLDRUSH - Move the miner around the different levels of the mine collecting gold and dodging high explosive charges and falling rocks. The green ledges crumble if you don't get off quickly. When you have collected all of the gold, get to the lift cage for a new screen.

PYRAMID - This is another fascinating text adventure built around the vaults of a pyramid.



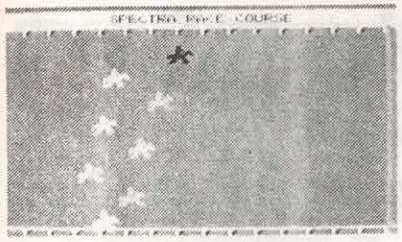
SVPAK25 - MSXPAK25 -

OZMAP - You must drive the car around Australia finding the places required as quickly as you can.

OZQUIZ - Answer general knowledge questions about Australia and her people.

MATH DUEL - Makes practicing maths fun! Has 4 levels of addition and subtraction maths.

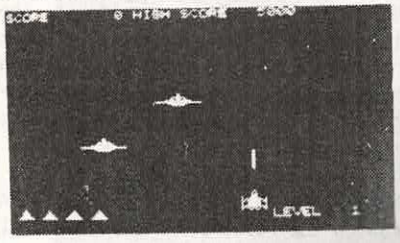
Prices include postage and packing within Australia



SVPAK1 - MSXPAK1 -

SPECTRA DERBY - Place your bets and have a day at the races. Even if you lose its painless.

HANGMAN - The classic word guessing game in which you will be gradually hung for each wrong letter.

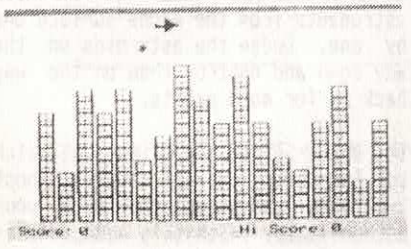


SVPAK10 - MSXPAK10 -

DEMON ATTACK - Shoot the attacking demons as they swoop overhead. You can only destroy a demon when it is fully formed. Killing ten demons gets you on to a harder set.

SEA DIVER - Dodge sea creatures as you make your way to the sea bed. Grab a gold bar and get back to the surface.

DEATH CITY - Race around the maze trying to shoot your computer opponent but be careful, he is also shooting at you.



SVPAK2 - MSXPAK2 -

SNAKE - Move the snake around to eat the diamonds, but mind you do not cross your own tail.

CITY BOMBER - You must destroy the towering buildings of the city before you can land safely.

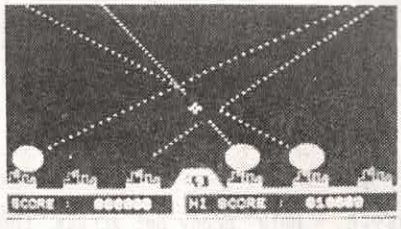


SVPAK8 - MSXPAK8 -

EXPLORER - Watch out for the Pitfalls in this game which will test your control of the joystick. Crocs and other obstacles bar your way as you wander through the jungle in search of treasures.

PARACHUTE - Catch the paratroopers in your boat as they jump from a helicopter into a shark-infested lagoon. The more you catch, the faster they jump.

ANDROID CAVERNS - Move around the caverns of this game and collect the treasure guarded by the androids. But watch out, the androids learn by their mistakes.

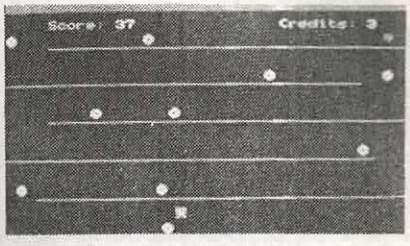


SVPAK22 - MSXPAK22 -

MISSILE COMMAND - Shoot down the incoming warheads before they destroy your cities. You get bonus cities awarded for high scores.

MASTERMIND - Your task is to match the same hidden four-colour combination that the computer has chosen.

LEAKY ROOF - Catch the raindrops leaking through the roof and empty the bucket out of the window before you drown.

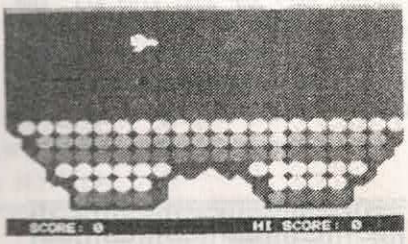


SVPAK4 - MSXPAK4 -

SPECTRA KONG - Win the lady's heart by getting to the top of the screen - but watch out for falling barrels.

TENNIS - A colour version of the original black and white TV game.

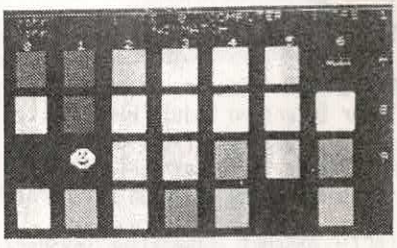
MUSICAL KEYBOARD - This program turns your computer keyboard into a musical keyboard and allows you to select note length and octave.



SVPAK3 - MSXPAK3 -

BOMBS AWAY - Destroy the targets in the ravine below. Be warned - it gets harder the more targets you hit.

BUG CATCHER - Two players move around the garden and the one with the most bugs caught - wins. Great for younger children.

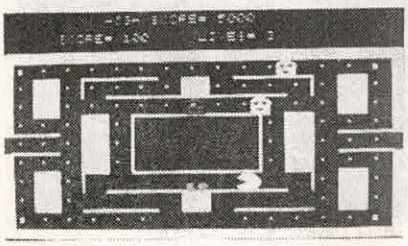


SVEP1 - MSXEP1 -

COPY CAT - The computer will play a set of notes and you must play them back in sequence. There are two levels of play - one note added to the same sequence or a new sequence each time.

MEMORY - A 1 or 2 player game of memory exercise. Uncover two squares on the screen and if the objects revealed match you get another chance

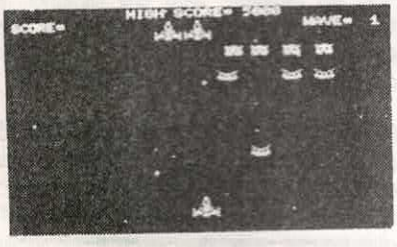
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SVPAK12 - MSXPAK12 -

MUNCH MAN - Guide your Munchman around the maze eating the dots while avoiding the deadly ghosts. If you eat one of the larger dots you will be able to eat the ghosts for a short while. Gain extra points by eating the fruit that occasionally appears on the screen.

ESCAPE FROM PLANET BETA - This is a text adventure where you are stranded on Planet Beta and must find all the necessary equipment to get off before the planet explodes. Only requires one or two-word commands - eg GET LASER.



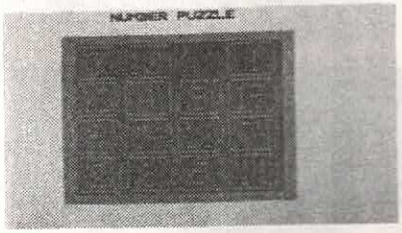
SVPAK5 - MSXPAK5 -

LUNAR LANDER - You are in command of a Lunar Excursion Module with limited fuel but full rotational ability and must land on the lunar surface. Has 8 different landscapes which get harder

GALACTIC ASSAULT - The alien invaders hover above you and one at a time dive on you. Destroy each wave before they get you. For 1 or 2 players.

ROAD RACER - Try and pass as many cars just as fast as you can without crashing.

SPRITE DEFINER - Design sprite shapes on the screen and rotate them when you have finished. Has grids for both sprite sizes.



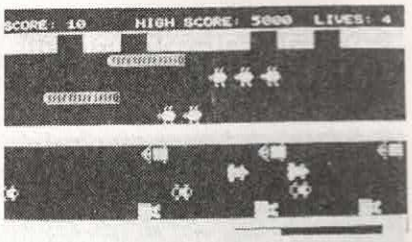
SVPAK20 - MSXPAK20 -

ARTILLERY FIRE - A two player game where each can select a firing angle and velocity to lob shells on his opponent.

NUMBER PUZZLE - Computerised version of the popular hand-held game in which you must slide numbers around into ascending order.

TOWERS OF HANOI - Move all of the discs from peg 1 to peg 3 without placing a large disc on a smaller one.

TENNIS - This is a two player game of Table Tennis.



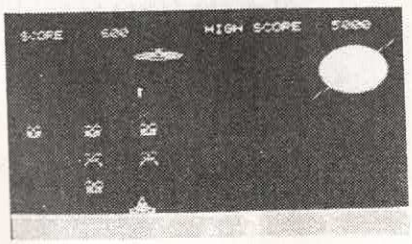
SVPAK6 - MSXPAK6 -

HOPPER - Help Hoppy get across the road (SPLAT!) and then to cross the river (SPLOSH!) and if you are lucky he won't get eaten by the crocs (GULP!!).

SPACE WAR - Challenging game for two players in this duel between your spaceship and that of your opponent. First one to ten points wins.

OTHELLO CHALLENGER - The classic mind game where you must outflank your opponent to capture his pieces. For 1 or 2 players.

PAINTER - Explore the flexibility of your computers colours with this on-screen paint brush.

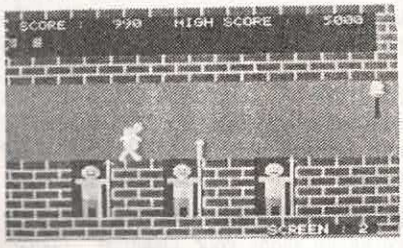


SVPAK9 - MSXPAK9 -

SPACE INVADERS - Shoot the aliens before they invade your planet. Move your ship left and right and shoot missiles at the aliens. Once they have landed the game will end. Score extra points by shooting the UFO which flies overhead.

SUB HUNT - Drop depth charges on the attacking subs and dodge the missiles they fire. You can fire two depth charges at once by firing while moving left or right. Each of the subs is marked with its value.

CHICKEN - Who cares why the chicken crossed the road, we have to find out how! Each time you make it across, the cars speed up.



SVPAK11 - MSXPAK11 -

LUNAR RESCUE - Rescue the stranded astronauts from the moons surface one by one. Dodge the asteroids on the way down and destroy them on the way back up for more points.

SKY BASE - Blast the aliens but watch out for their kamikazi attacks. Shoot two sets and then dock with your mother ship. With this added fire-power go on to destroy the alien base.

QUASIMODO - Can you as the hunchback Quasimodo rescue the beautiful maiden from the French soldiers. Cross the screen avoiding fireballs and spears, then ring the bell to get the next screen.

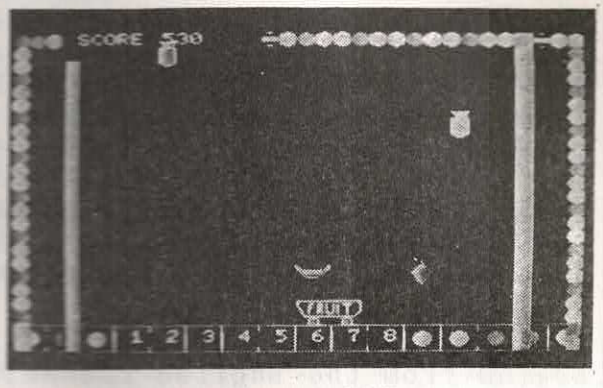
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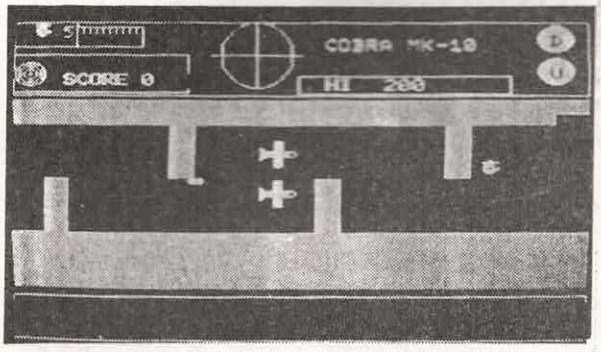
SVPAK26 - MSXPAK26 -

JUMPING COWBOY - A 'platform' game of nine screens where you must collect all the opals to get the next screen.

SUPER SMASHER - Move the bat to smash the ball through the two brick walls. Has three levels of difficulty.

FRUIT CATCHER - Catch falling fruit in your cart. As you catch more, the types of fruit will change and fall at a faster rate.





SVPAK27 - MSXPAK27 -

MIRKWOOD FOREST - Graphics adventure where you must escape from the forest by finding four keys, eating food to maintain energy, while avoiding the spiders. This one will really test your adventure gaming ability.

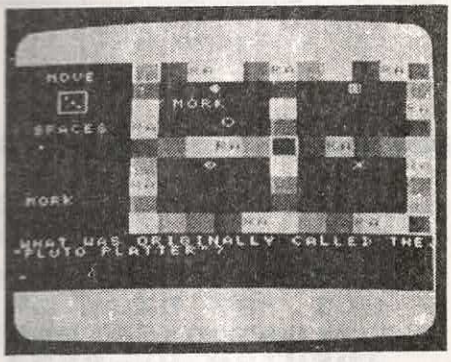
CHOPPER PILOT - The fourteen levels of this game make it very compelling. Keeping clear of canyon walls, fly your helicopter to do battle with enemy fighters, bombers, missiles and UFOs. Great skill required.

UFO (MSX ONLY)

In this machine code game, you are under attack from UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'. Watch out for the meteor shower. Refuel when the truck appears on screen.
Requires 16K RAM, has smooth scrolling, fast response with good sound effects.

METEOR SWARM (SV ONLY)

Another machine code game where your ship is caught in a meteor belt and you must blast your way out. Other ships are caught in the belt and will shoot at you if you do not destroy them first. Comes with bonus game - Birds of Orion.



SPECTRA-TRIVIA (SV 32K)
COMPU-TRIVIA (MSX)

The famous trivia game now on your computer. These are trivia quiz games for up to four players. The computer rolls the dice and each player moves in turn. At certain points you will be asked questions. There is provision for you to create your own question and answer file.

PLEASE SPECIFY SV318/328 OR MSX WITH ORDER.

========== D A M B U S T E R S ==========

There are many Flight Simulators on the market for MSX but few are as good or have as many appealing qualities as DAMBUSTERS. The program is a simulation of squadron 617's attempt to blow up three dams in order to cause havoc in the heart of wartime Germany.

Many of the difficulties overcome by squadron 617 will also have to be negotiated by yourself, you must take control of all the crew members jobs from the PILOT to ENGINEER, no mean feat!

There are 2 levels of play (1) As Flight Lieutenant your flight begins from the English channel. (2) As Squadron Leader you takeoff from Scampton airfield the date is MARCH 21st 1943. You are flying a Lancaster Bomber.

Once over the English channel you will encounter ME109 fighters, searchlights from military installations, plus various other hazards. Eventually, applying good navigation & piloting skills you will get your shot at bombing the dams at DORTMUND, DUISBURG & SOEST, a most daring task, height, speed & distance all have to be very precise to achieve a direct hit.

The game is well made & fairly addictive with excellent graphics plus atmspheric Chugs & Bangs.

Excellent value at $25.00 Cassette or Disk from -

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PROGRAMMING WITH SPRITES - PART 3.

By Lucille Parker

SPRITE PLANES.

You have probably noticed that sprites aLways pass in front of anything else drawn on the screen. This is because they are on the sprite planes (surface), and each one must be on a different plane. There are 32 planes available, numbered 0 to 31, which means that the maximum number of sprites you could have on the screen at one time is 32. You can actually have up to 256 small sprites in memory in a program or 64 large ones, space permitting.

An important rule to remember when using sprites in a program is that you can't have more than 4 sprites in a row horizontally, although you can have more vertically. You won't get an error message if you end up with 5 or more across a row, they will just disappear, or partly disappear. Invisible sprites won't register if they collide.

The following program demonstrates how you can put a sprite on the screen at 32 different locations, each on a different plane. The first number in the PUT SPRITE statement is the sprite plane number, and the last number is the SPRITE$ number. Be careful not to confuse these.

```
10 SCREEN1,0: REM SCREEN 2,0 FOR MSX
20 SPRITE$(1)="BBFF~~<$"
30 X=50:Y=1:A=0:B=3
40 FOR T=1TO8:FOR L=ATOB
50 PUT SPRITE L,(X,Y),15,1
60 X=X+50:NEXT L
70 A=A+4:B=B+4:X=50:Y=Y+25:NEXT T
80 GOTO 80
```

Sprites pass in front of one another according to the priority of their plane numbers. The sprite on plane 0 has top priority and will pass in front of all the others. Order of priority goes back through the planes to the sprite on plane 31, which has the lowest priority.

Just for fun, I have set up the following demo program so that sprite 2 remains on plane 2, but sprite 1 passes alternately in front of and behind it, on each pass across the screen. The variable P represents the plane number for sprite 1. Line 90 switches this value from 1 to 3 or 3 to 1 when the sprites reach the edge of the screen.

```
10 COLOR 15,5,15:SCREEN1,2:REM SCREEN 2,2 FOR MSX
20 FOR T=1TO2:FOR U=1TO32:READ A:A$=A$ + CHR$(A):NEXT:SPRITE$
(T) = A$:A$="":NEXT
30 DATA0,0,0,0,31,63,31,0,0,0,0,0,0,0,0,0,5,10,30,62,254,254,
254,62,30,10,5,0,0,0,0
40 DATA0,4,14,14,14,14,14,14,14,14,31,63,127,191,127,128,0,0,
0,0,0,0,0,0,0,0,128,192,160,192,32
50 P=1
60 X1=250:Y2=192
70 PUT SPRITE P,(X1,104),1,1:X1=X1-2
80 PUT SPRITE 2,(128,Y2),15,2:Y2=Y2-1.4
90 IFX1<-14 THEN IF P=1 THEN P=3 ELSE P=1
100 IFX1<-14 THEN60
110 GOTO 70
```

LODE RUNNER

Load runner would have to be one of the best games that I have played since PAYLOAD, it has very good graphics and is the down and let the hours pass by.

This game has some very good features, such as, being able to pick any level you like to play on, and for those who hate loosing, you can increase you lives up to 99 if you wish.
You can change the sound by turning it on or off.
Change the speed so your man and the ones chasing you go faster.

You must be able to collect the gold bricks on each level to be able to go up to the next level, sounds pretty easy, well actually the first few levels are but then it takes a bit of thinking to get up to the next level and so on.

It is a very addictive game, so be prepard for late nights.

The disk version has even more features, it let's you create your own screens and copy them onto disk for use at any time.

OVERALL COMMENTS -

GRAPHICS - V/GOOD
SOUND EFFECTS - GOOD
EXCITEMENT - V/GOOD
VALUE FOR MONEY- EXCELLENT



BATTLE CROSS

Battle Cross is another outer space tpe game where you shoot down aliens, which look a lot like stingrays. The game is played by either Joystick or cursor controls for 1 or 2 players.

You start off above a planet and shoot down aliens that pass you and come at you from their mother ship.

The only catch being that the planet gives off an electrical charge and if it hits you, you fry.

On level 2 you must maneuver your ship through an asteroid field with out hitting anything, and while you doing this you must still shoot down those annoying stingrays (aliens).

I personally did not think it was all that good but remember that is my opinion, and everyone has different taste and views.
So I leave it up to you to judge for yourself.

OVERALL COMMENTS -

GRAPHICS - GOOD
SOUND EFFECTS - AVERAGE
EXCITEMENT - AVERAGE
VALUE FOR MONEY- FAIR



USER GROUPS

SYDNEY

We have been advised of a group in Sydney for Spectravideo owners- Please contact Malcolm Samuels on 888 7664 after hours.

QUEENSLAND

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We meet on the first Saturday afternoon of each month (except January) at the Nunawading Civic Centre 12.00 - 5.00

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Turka Finland

If any of the above groups wish to enter their meeting times & place or phone number, drop us a line and we will include them in this page.

BATTLE

Here is a little game for 2 Players

use Cursor Keys or joysticks to move your men around.
This listing was supplied by:-
MSX INTERNATIONAL

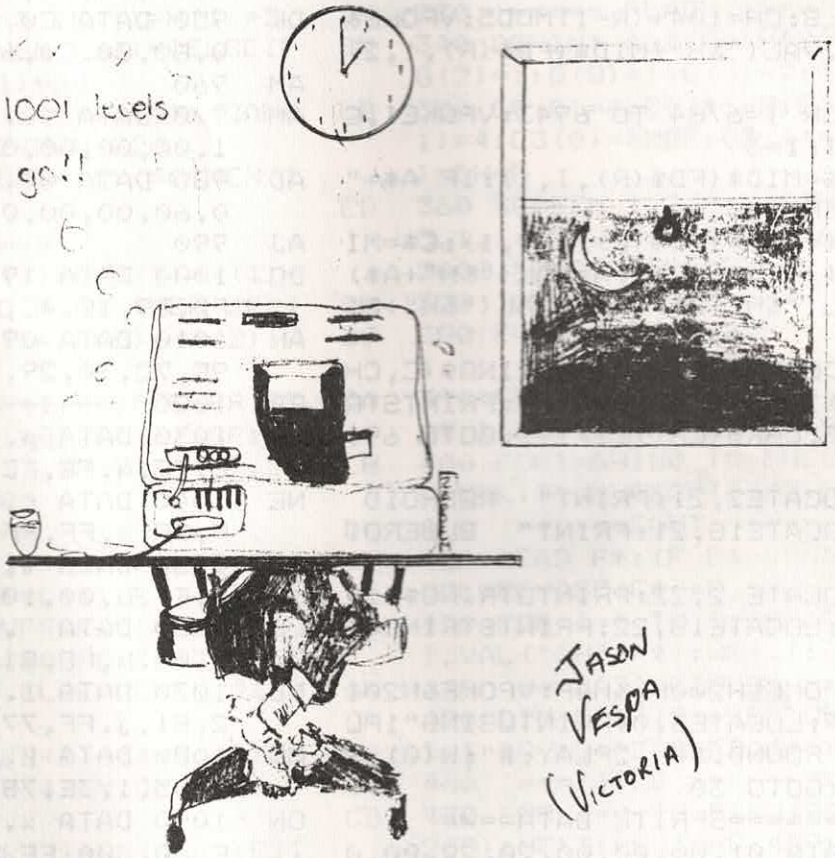
```
CN  10 GOTO 340
EP  30 FOR I=0 TO 1:A=STICK(I+K):Q=
    P+X(I)+Y(I)*32+64
DJ  40 '====JUMP====
JF  45 IF G(A)>1 THEN H(I)=G(A)-3
DF  50 IF VPEEK(Q)=CH OR VPEEK(Q+1)
    =CH THEN 140
GE  60 Y(I)=Y(I)-SGN(JF(I)):X(I)=X(
    I)+JP(I):JF(I)=JF(I)-1:IF X(I)<
    0 OR X(I)>30 THEN JP(I)=-JP(I):
    H(I)=-H(I)
DD  70 IF Y(I)<2 THEN JF(I)=-1
GG  80 SP=5-H(I):KF(I)=1:PUTSPRITEI
    *2,(X(I)*8,Y(I)*8),C0(I),SP:PUT
    SPRITEI*2+1,(X(I)*8,Y(I)*8),C1(
    I),SP+1
CE  90 IF E(I)THEN 180ELSE Q=P+X(I)
    +Y(I)*32:IF STRIG(I+K)ANDVPEEK(
    Q)<>CHANDVPEEK(Q+1)<>CH THEN SO
    UND7,&HFE:PLAY"T255V10L6407C":C
    (I)=X(I):D(I)=Y(I):E(I)=H(I)*2
JG  100 IF X(I)<0 THEN X(I)=0ELSEIF
     X(I)>30THEN X(I)=30
BA  110 NEXT:IF ABS(X(0)-X(1))<2AND
    ABS(Y(0)-Y(1))<2THEN 220
FI  120 IF BP(0)<=0ORBP(1)<=0THEN 1
    130ELSE30
FD  130 '====MOVE=====
OA  140 JF(I)=-1:JP(I)=0:X(I)=X(I)-
    (A=3)+(A=7):IF G(A)=1 THEN JF(I
    )=8:JP(I)=0:JP(I)=(A=8)-(A=2):G
    OTO 60
MM  160 SP=1+H(I):KF(I)=0:PUTSPRITE
    I*2,(X(I)*8,Y(I)*8),C0(I),SP:PU
    TSPRITEI*2+1,(X(I)*8,Y(I)*8),C1
    (I),SP+1:GOTO 90
AM  170 '====BEAM====
BH  180 Q=P+C(I)+D(I)*32:VPOKEQ,32:
    VPOKEQ+1,32:C(I)=C(I)-E(I):Q=P+
    C(I)+D(I)*32:IFVPEEK(Q)=CHORVPE
    EK(Q+1)=CHORC(I)<0ORC(I)>30THEN
    E(I)=0:GOTO 100 ELSEVPOKEQ,96:V
    POKEQ+1,96
KC  190 J=1+(I=1):IFC(I)<(X(J)-1)OR
    C(I)>(X(J)+1)ORD(I)<Y(J)ORD(I)>
    (Y(J)+1)THEN100
AE  200 GOSUB 280:BP(J)=BP(J)-1:E(I
    )=0:VPOKEQ,32:VPOKEQ+1,32:GOSUB
    300:GOTO 100
EH  210 '====================
HH  220 SOUND8,&H10:SOUND9,&H10:SOU
    ND11,&H5:SOUND12,&H9:SOUND13,&H
    0:SOUND0,&HAA:SOUND1,&H0:SOUND2
    ,&H96:SOUND3,&H0:SOUND7,&HF7
CB  230 J=-1:IF KF(0)+KF(1)=1 THEN
    J=0:IF KF(0)=1 THEN J=1
GN  240 IF J<>-1 THEN BP(J)=BP(J)-1
    :PUTSPRITEJ*2,(X(J)*8,Y(J)*8),1
    5,5-H(J):PUTSPRITEJ*2+1,(X(J)*8
    ,Y(J)*8),15,6-H(J):GOSUB 300:GO
    SUB 280
AN  250 A=SGN(X(0)-X(1)):X(0)=X(0)+
    A:X(1)=X(1)-A:IF A=0 THEN A=H(0
    )
HA  260 JP(0)=A:JP(1)=-A:JF(0)=-JF(
    0):JF(1)=-JF(1):GOTO 120
DC  270 '===== SOUND SUB=======
IO  280 SOUND6,&HF:SOUND8,&H10:SOUN
    D11,&H0:SOUND 12,&H8:SOUND13,&H
    0:SOUND0,&H14:SOUND1,&H5F:SOUND
    7,&HF6:RETURN
DC  290 '===== PRINT SUB
AP  300 IF BP(0)<12 THEN LOCATE 13-
    BP(0),22:PRINT" "
BD  310 IF BP(1)<12 THEN LOCATE 29-
    BP(1),22:PRINT" "
JH  320 VPOKE&H200F,C3(BP(0)/5):VPO
    KE&H2010,C3(BP(1)/5):RETURN
BN  330 '===== READY ====
BM  340 DEFINT A-Z:P=&H1800:G(1)=1:
    G(2)=1:G(8)=1:G(3)=2:G(7)=4
LB  350 C0(0)=4:C0(1)=8:C1(0)=8:C1(
    1)=4:C3(0)=&H8F:C3(1)=&HBF:C3(2
    )=&H4F
ED  360 SCREEN 1,2,0:WIDTH32:COLOR1
    5,1,1:KEYOFF:RESTORE760:VPOKE&H
    200C,&HB0:VPOKE&H200E,&H40
CH  370 '=====SPRITE ====
BJ  380 FORJ=0 TO 8:B$="":FORI=0TO3
    1
CA  390 READA$:B$=B$+CHR$(VAL("&H"+
    A$)):NEXT:SPRITE$(J)=B$:NEXT
LH  400 FORI=&H180 TO &H2D7:VPOKEI,
    VPEEK(I)ORVPEEK(I)/2:NEXT
DK  410 '======FONT =====
BL  420 READ F$:IF F$="END" THEN 45
    0ELSEF=ASC(F$)*8
FH  430 FOR I=0 TO 7:READF$:VPOKEF+
    I,VAL("&H"+F$):NEXTI:GOTO 420
AK  440 '===DATA FIELD ====
CM  450 DIM FD$(12):RESTORE1290:FOR
    I=1 TO 12:READ FD$(I):NEXT
BE  460 '===TITLE =========
CD  470 FOR I=0 TO 5:PUTSPRITEI,(0,
    209):NEXT:RESTORE 480:CLS:FORI=
    2 TO 21:READI$:LOCATE3,I:PRINTI
    $:NEXT
```

```
BE  480 DATA"          pq          pq "
BB  490 DATA"        pq     pq pqpq   "
BP  500 DATA"      pssrsrssrssrrssq"
EE  510 DATA"     pqprrqpq pqprstq "
HE  520 DATA"   pssrssrsqpsrrsttq",
EF  530 DATA"          pssssq        ps
    q"
EG  540 DATA"        psq pq    psq psq
    "
BN  550 DATA"      psssrssssrtrssssq"
BM  560 DATA"     psqpqpsqprsrqpsq"
BL  570 DATA"    psqpsrssssrsrssssq",,
BH  580 DATA"                       "
BG  590 DATA"
     ",,
CH  600 DATA"<1>KEY:CURSOR & STICK1
    ",
BL  610 DATA"<2>KEY:STICK1 & STICK2
    "
BB  620 R=1:W(0)=0:W(1)=0
MA  630 I$=INKEY$:IF I$="1" OR I$="
    2" THEN K=VAL(I$)-1 ELSE 630
DI  640 '-----
PI  650 X(0)=2:Y(0)=18:JP(0)=0:E(0)
    =0:BP(0)=12:H(0)=-1:X(1)=28:Y(1
    )=Y(0):JP(0)=0:E(1)=0:BP(1)=BP(
    0):H(1)=1
DG  660 '-------
EN  670 CLS:CH=104+(R-1)MOD5:VPOKE&
    H200D,VAL("&H"+MID$(FD$(R),1,2)
    ):I=3
CM  680 FOR I=6784 TO 6943:VPOKEI,C
    H:NEXT:I=3
HO  690 A$=MID$(FD$(R),I,1):IF A$="
    X" THEN 720
AO  700 B$=MID$(FD$(R),I+1,1):C$=MI
    D$(FD$(R),I+2,1):A=VAL("&H"+A$)
    :B=VAL("&H"+B$)+2:C=VAL("&H"+C$
    )
ED  710 LOCATEA,B:PRINTSTRING$(C,CH
    R$(CH)):LOCATE32-A-C,B:PRINTSTR
    ING$(C,CHR$(CH)):I=I+3:GOTO 690

BP  720 LOCATE2,21:PRINT"  REDROID
      ":LOCATE18,21:PRINT"  BLUEROI
    D  "
DA  730 LOCATE 2,22:PRINTSTRING$(12
    ,"x"):LOCATE18,22:PRINTSTRING$(
    12,"")
FP  740 VPOKE&H200F,&H4F:VPOKE&H201
    0,&H4F:LOCATE3,0:PRINTUSING"1PL
    AY:#  ROUND:##  2PLAY:#";W(0);R
    ;W(1):GOTO 30
DM  750 '======SPRITE DATA====
AP  760 DATA 01,00,00,00,20,20,00,0
    0,01,01,01,01,01,01,01,00
DH  770 DATA C0,00,08,0C,04,0C,08,8
    0,80,00,00,80,80,C0,C0,00
AI  780 '
BB  790 DATA 00,03,0F,1F,1F,1F,07,0
    1,00,00,00,00,00,00,00,01
BD  800 DATA 00,E0,F0,F0,F8,F0,F0,7
    0,60,00,00,00,00,00,00,E0
BA  810 '
AH  820 DATA 03,00,10,30,20,30,10,0
    1,01,00,00,01,01,03,03,00
BF  830 DATA 00,00,00,00,04,04,00,0
    0,80,80,80,80,80,80,80,00
AN  840 '
BH  850 DATA 00,07,0F,0F,1F,0F,0F,0
    E,06,00,00,00,00,00,00,07
AK  860 DATA 00,C0,F0,F8,F8,F8,E0,8
    0,00,00,00,00,00,00,00,80
AK  870 '
AN  880 DATA 03,00,10,30,20,30,10,0
    1,01,01,03,06,0E,0E,04,00
BL  890 DATA 80,00,00,00,04,04,00,0
    0,00,00,00,00,00,00,00,00
BC  900 '
AD  910 DATA 00,07,0F,0F,1F,0F,0F,0
    E,06,00,00,20,30,10,08,00
AN  920 DATA 00,C0,F0,F8,F8,F8,E0,8
    0,00,00,00,00,00,00,00,00
AP  930 '
AM  940 DATA 01,00,00,00,20,20,00,0
    0,00,00,00,00,00,00,00,00
DK  950 DATA C0,00,08,0C,04,0C,08,8
    0,80,80,C0,60,70,70,20,00
AM  960 '
AM  970 DATA 00,03,0F,1F,1F,1F,07,0
    1,00,00,00,00,00,00,00,00
AD  980 DATA 00,E0,F0,F0,F8,F0,F0,7
    0,60,00,00,04,0C,08,10,00
AJ  990 '
DO 1000 DATA 19,45,63,3F,9B,4D,1E,
    FD,39,1E,4C,97,BE,33,63,CA
AK 1010 DATA 09,23,66,AC,FD,B8,74,
    9F,7C,30,29,F1,FA,98,2C,62
EB 1020 '-------- FONTS ---------
MP 1030 DATA p,00,01,03,07,0F,1F,3
    F,7F,q,FE,FC,F8,F0,E0,C0,80,00
NE 1040 DATA r,FE,FD,FB,F7,EF,DF,B
    F,7F,s,FF,FF,FF,FF,FF,FF,FF,FF
ON 1050 DATA t,00,FF,FF,FF,FF,FF,F
    F,FF,u,00,10,FE,7C,38,6C,00,00
LE 1060 DATA `,00,00,00,00,FF,00,0
    0,00,h,FF,81,FD,FD,FD,FD,FD,FF
MD 1070 DATA i,FF,BE,DC,E8,F8,E4,C
    2,81,j,FF,77,77,77,33,33,11,11
MK 1080 DATA k,FF,7F,BF,5F,AF,57,A
    B,D5,l,3E,7B,7F,7D,7D,7D,7F,7F
ON 1090 DATA x,00,FF,FF,FF,FF,FF,F
    F,00,,00,FF,FF,FF,FF,FF,FF,00
DN 1100 DATA ',FF,00,00,00,00,00,0
    0,FF
CA 1110 DATA "END"
```

```
DB  1120 '---------- OUT!--------
DI  1130 PLAY"T255S0M9000V1501":SOU
    ND7,&HF2:SOUND6,18:PLAY"GEEDC"
CF  1140 FOR J=0 TO1:IF BP(J)>0 THE
    N W(J)=W(J)+1ELSE PUTSPRITEJ*2,
    (X(J)*8,Y(J)*8),11,8
AK  1160 NEXT:FORI=0 TO 4000:NEXT
AN  1170 IF W(0)<7AND W(1)<7 THEN R
    =R-(R<12):GOTO 650
DM  1180 CLS:J=-(W(0)<W(1))
EJ  1190 FOR I=0 TO 1000:NEXT
MC  1200 LOCATE 7,3:PRINT"uuu  TOTA
    L uuu
HJ  1210 LOCATE 6,5:PRINT"REDROID
     BLUEROID
CA  1220 LOCATE10,7:PRINTUSING"##
    -  ##";W(0);W(1)
GF  1230 LOCATE9,15:PRINT"PLAYER";J
    +1;" WIN"
AI  1240 LOCATE6,17:PRINT"uuG A M E
     O V E Ruu"
BP  1250 LOCATE8,19:PRINT"HIT RETUR
    N KEY"
FN  1260 PUTSPRITE0,(64,80),4,0:PUT
    SPRITE1,(64,80),8,1:PUTSPRITE2,
    (160,80),8,2:PUTSPRITE3,(160,80
    ),4,3
CP  1270 IF INKEY$=CHR$(13)THEN 470
    ELSE 1270
AI  1280 '-------- FIELD DATA ------
ND  1290 DATA E0945384BA55D4X,D0654
    A823C5ED2X
DN  1300 DATA F4551C512918A1F915D1B
    D1X,A0034463793AC3C44X
BM  1310 DATA 60934D623839D4X,20532
    A32E32582A82E825D2AD2ED2X
HD  1320 DATA 4EC41971D916A1AC13D1X
    ,A0D42354B835B4BD3X
FD  1330 DATA A0227E32275BA32C7X,E0
    A04634474E726B4AE4X
GB  1340 DATA D4D314549846B1DD1X,F4
    233A465832C3AC6X
CK  1350 '==========================
    =====
BL  1360 ' BATTLE ROID
CM  1370 '==========================
    =====
```

THE END



JOUST

For SV318/328 Computers.

This program was supplied by Ross Moore. It is based on the old Atari/Microbee game of the same name. HAPPY HUNTING!!

```
DF  10 ONSTOPGOSUB960:STOPON:GOTO97
    0
HL  20 COLOR,1,1:SCREEN1,2:DEFINTA-
    C,E-Z:DEFSNGD:CLICKOFF
DC  30 PLAY"T255L64":SOUND6,31:SOUN
    D7,&B101010:B$="T255L64V15O7ED#
    DV12ED#DV8ED#D":K$="V1406CER6CG
    3"
EA  40 STOPON:GOSUB790:COLOR15
JN  50 CLS:STOPON:COLOR15:LINE(0,17
    0)-(255,191),13,BF:LOCATE110,18
    3:PRINT"Hi:"HS:LOCATE198,183:PR
    INT"Lives:":S=3:X=120:Y=153:A=2
    40:B=90:DA=-5:DB=1.8:N=RND(-TIM
    E):S1=1:S2=4:S3=3:LI=5:GOSUB640
    :GOSUB780:C=10
EC  60 ONSPRITEGOSUB370:GOTO180
AC  70 IFDY>-2THENDY=DY-5ELSEDY=DY-
    2:IFDY<-7THENDY=-7
BI  80 RETURN
BA  90 IFPOINT(X+17,Y+4)=13ORPOINT(
    X+17,Y+11)=13THENPLAYB$:DX=-DX
CB  100 RETURN
CJ  110 IFPOINT(X-2,Y+4)=13ORPOINT(
    X-2,Y+11)=13THENPLAYB$:DX=-DX
BP  120 RETURN
EI  130 IFPOINT(X+8,Y+22)=13THENDY=
    1
DA  140 IFPOINT(X+8,Y+17)=13THENDY=
    0:IFDX<0THENS1=2:RETURN300ELSES
    1=1:RETURN300
CD  150 IFPOINT(X+8,Y+13)<>8ANDPOIN
    T(X+8,Y+9)<>8THENRETURNELSESOUN
    D9,15:PUTSPRITE0,,15,8:FORN!=0T
    O31STEP.05:SOUND6,N!:NEXT
GD  160 FORN=0TO4:SOUND9,12-3*N:PUT
    SPRITE0,,,N+9:FORT=1TO100:NEXTT
    ,N
EP  170 LI=LI-1:GOSUB780:X=120:Y=15
    3:DX=0:DY=0:RETURN :' *****
KE  180 PUTSPRITE0,(X,Y),15,S1:PUTS
    PRITE1,(A,B),C,S
DC  190 A=A+DA:B=B+DB:IFB>153THENB=
    153:DB=-DBELSEIFB<0THENB=0:DB=-
    DB
AG  200 ST=STICK(0)+STICK(1):SR=STR
    IG(0)+STRIG(1)
CN  210 IFST=2ORST=3ORST=4THENDX=DX
    +1
CA  220 IFST=6ORST=7ORST=8THENDX=DX
    -1
DO  230 IFDX>7THENDX=7ELSEIFDX<-7TH
    ENDX=-7
AO  240 IFS=5ANDTIME>600THENGOSUB54
    0
AL  250 IFSRTHENGOSUB70
DK  260 DY=DY+1:IFDY>7THENDY=7
CC  270 IFDY>0THENGOSUB130
AI  280 IFDX>0THENGOSUB90ELSEIFDX<0
    THENGOSUB110
AE  290 IFDX>=0THENS1=S2ELSES1=S3
BA  300 X=X+DX:Y=Y+DY:IFY<0THENY=0:
    DY=1
CL  310 IFX<0THENX=240ELSEIFX>240TH
    ENX=0
CN  320 IFRND(1)<5E-03THENDA=-DA:IF
    S<5THENS=7-SELSEIFS>5THENS=13-S

BP  330 IFRND(1)<.01THENDB=-DB
PI  340 IFDY<=0THENIFPOINT(X+8,Y-2)
    =13THENDY=6
AE  350 SM=1-SM:IFSMTHENSOUND9,8ELS
    ESOUND9,4
BH  360 SPRITEON:S=10-S:S2=10-S2:S3
    =10-S3:GOTO180:'*****
BN  370 SPRITEOFF:SOUND9,0:IFS=5THE
    N520ELSEIFX=120ANDY=153THENRETU
    RN
DI  380 IFY>B-3ANDY<B+3THEN440
CF  390 IFSGN(DA)=SGN(DX)THENPLAYB$
    :GOTO460
AF  400 IFY>BTHEN410ELSEPLAYK$:S=5:
    SC=SC+150:B=B+20:DY=DY-3:DB=1.8
    :GOSUB640:TIME=0:RETURN180
AF  410 SOUND5,3:SOUND10,14
EO  420 FORD=YTO238:PUTSPRITE0,(X,D
    ),,8:SOUND4,D:NEXT:SOUND10,0
BC  430 LI=LI-1:S1=1:Y=153:X=120:DX
    =0:DY=0:GOSUB780:RETURN180
CG  440 PLAYB$:IFSGN(DA)=SGN(DX)THE
    N460
CI  450 DA=-DA:DX=-DX:GOTO480
AD  460 IFABS(DA)>ABS(DX)THENDA=-DA
    :DX=2*DX:GOTO480
CL  470 DX=-2*DX:GOTO490
BA  480 IFS<5THENS=7-SELSEIFS>5THEN
    S=13-S
CG  490 X=X+2*DX:A=A+2*DA
CN  500 IFB>YTHENB=B+8ELSEB=B-8
AF  510 RETURN180
DK  520 A1=A:B1=B:SC=SC+300:B=RND(1
    )*150:PLAY"O7V14GG+AV15A+BO8C":
    IFX<120THENA=240:DA=-5:S=3ELSEA
    =0:DA=5:S=4
AF  530 GOTO550
EF  540 A1=A:B1=B:SC=SC-100:PLAY"S1
    1M800007B2":IFDA=5THENS=4ELSES=
    3
```

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AM  550 C=C-2:IFC=1THENC=10ELSEIFC=
    8THENC=7
CI  560 FORN=9TO13:PUTSPRITE1,(A1,B
    1),N-5,N:FORT=1TO50:NEXTT,N
BG  570 GOSUB640:RETURN180:'*****
BL  580 C=15:PUTSPRITE0,,0:PUTSPRIT
    E1,,0:IFSC>HSTHENHS=SC
KH  590 CLS:N1=43:PLAY"V14":FORN=1T
    06:N2=N1-3:N3=N1-6:N4=N1-9:PLAY
    "N=N1;N=N2;N=N3;N=N4;":N1=N1-3:
    NEXT:N4=N4-3:PLAY"L2N=N4;L64"
GB  600 IFSTRIG(0)+STRIG(1)THENSC=0
    :GOTO50
BD  610 C=C-1:IFC=1THENC=15
EB  620 COLORC:LOCATE73,20:PRINT" G
     A M E   O V E R ":LOCATE40,60:P
    RINT"You scored"SC:LOCATE40,80:
    PRINT"High score:"HS
BM  630 LOCATE40,150:PRINT"PRESS 'F
    IRE' FOR NEW GAME.":GOTO600
DA  640 LINE(26,183)-(56,190),13,BF
    :COLOR15:LOCATE20,183:PRINTSC:O
    NSC/500 GOSUB650,660,670,680,69
    0,700,710,720,730,740,750,760:R
    ETURN
AE  650 LINE(75,80)-(175,87),13,BF:
    RETURN
BI  660 MA=0:LINE(0,40)-(75,47),13,
    BF:LINE(175,40)-(255,47),13,BF:
    RETURN
BO  670 LINE(0,118)-(75,125),13,BF:
    LINE(175,118)-(255,125),13,BF:R
    ETURN
CI  680 LINE(0,170)-(50,180),8,BF:L
    INE(205,170)-(255,180),8,BF:RET
    URN
BB  690 LINE(50,170)-(100,180),8,BF
    :LINE(155,170)-(205,180),8,BF:R
    ETURN
BL  700 LINE(100,170)-(155,180),8,B
    F:LINE(125,170)-(130,173),13,BF
    :IFMA=0THENGOSUB770:MA=1:RETURN
    ELSERETURN
DK  710 LINE(75,80)-(175,73),8,BF:R
    ETURN
AA  720 LINE(0,40)-(75,33),8,BF:LIN
    E(175,40)-(255,33),8,BF:RETURN
DP  730 LINE(0,118)-(75,111),8,BF:L
    INE(175,118)-(255,111),8,BF:RET
    URN
DO  740 LINE(0,0)-(7,170),13,BF:LIN
    E(255,0)-(248,170),13,BF:IFMA=1
    THENGOSUB770:MA=2:RETURNELSERET
    URN
BB  750 LINE(122,35)-(128,80),13,BF
    :RETURN
CC  760 LINE(0,0)-(255,6),13,BF:IFM
    A=2THENGOSUB770:MA=3:RETURNELSE
    RETURN
IN  770 LI=LI+1:SOUND5,0:SOUND10,14
    :FORN=1TO3:FORD=255TO0STEP-8:SO
    UND4,D:NEXT:FORD=5TO255STEP8:SO
    UND4,D:NEXTD,N:SOUND10,0
AN  780 LINE(240,183)-(245,190),13,
    BF:COLOR15:LOCATE234,183:PRINTL
    I:IFLI=0THEN580ELSERETURN
GA  790 RESTORE:FORS=1TO7:IFS=5THEN
    NEXTELSES$="":FORN=1TO32:READA:
    S$=S$+CHR$(A):NEXTN:SPRITE$(S)=
    S$:NEXTS
DK  800 S$="":FORN=1TO6:READA:S$=S$
    +CHR$(A):NEXT:SPRITE$(5)=S$
AN  810 S$="":FORN=1TO32:READA:S$=S
    $+CHR$(A):NEXT:SPRITE$(8)=S$
FD  820 FORS=9TO13:S$="":FORN=1TO29
    :READA:S$=S$+CHR$(A):NEXT:SPRIT
    E$(S)=S$:NEXT:RETURN
HC  830 DATA 6,6,4,14,14,14,127,31,
    127,255,127,31,2,2,2,3,0,0,6,15
    ,8,8,255,8,152,248,240,128,0,0,
    0,128:' Stand R
AG  840 DATA0,0,96,240,16,16,255,16
    ,25,31,15,1,0,0,0,1,96,96,32,11
    2,112,112,254,248,254,255,254,2
    48,64,64,64,192:'Stand L
CP  850 DATA0,0,96,240,16,16,255,16
    ,25,31,15,3,0,0,0,0,96,96,36,12
    4,124,124,254,252,254,255,254,2
    48,224,0,0,0:'Fly L(U)
AO  860 DATA6,6,4,14,14,14,127,31,1
    27,255,127,63,63,60,112,64,0,0,
    6,15,8,8,255,8,152,248,240,192,
    0,0,0,0:'Fly R(D)
CO  870 DATA6,6,36,62,62,62,127,63,
    127,255,127,31,7,0,0,0,0,0,6,15
    ,8,8,255,8,152,248,240,192,0,0,
    0,0:'Fly R(U)
BI  880 DATA0,0,96,240,16,16,255,16
    ,25,31,15,3,0,0,0,0,96,96,32,11
    2,112,112,254,248,254,255,254,2
    52,252,60,14,2:'Fly L(D)
GD  890 DATA60,126,255,255,126,60:'
    Egg
DP  900 DATA2,7,7,15,15,15,255,143,
    143,6,6,7,3,0,0,0,0,64,64,192,2
    48,255,251,192,64,64,64,64,248,
    76,76,72:' Dive
AL  910 DATA0,0,0,0,1,5,0,12,0,5,1,
    0,0,0,0,0,0,0,0,0,0,0,64,0,96,0,6
    4,0,0,0:'P sml
AI  920 DATA0,0,0,1,9,4,0,24,0,4,9,
    1,0,0,0,0,0,0,0,0,0,32,64,0,48,0,
    64,32,0,0:' *2
AE  930 DATA0,0,1,17,8,0,0,48,0,0,8
    ,17,1,0,0,0,0,0,0,0,16,32,0,0,24,
    0,0,32,16,0:'*3
```

```
AI  940 DATA0,1,1,16,8,0,0,96,0,0,8
    ,16,1,1,0,0,0,0,0,16,32,0,0,12,
    0,0,32,16,0:'*4
DC  950 DATA1,1,32,16,0,0,0,192,0,0
    ,0,16,32,1,1,0,0,0,8,16,0,0,0,6
    ,0,0,0,16,8:'*5
CL  960 IFINKEY$<>""THEN960ELSEBEEP:
    COLOR15,4:SCREEN0,1:CLICKON:END

BJ  970 DATA50,79,83,83,0,45,79,79,
    82,69,0,80,82,69,83,69,78,84,83
    ,14,14,14
DO  980 COLOR4,15:SCREEN0,0:LOCATE,
    ,0::DIML(63):RESTORE970:FORN=21
    TO42:READL(N):NEXT:PLAY"T200L30
    S8M2000":N1=30
FG  990 FORA=0TO20:PLAY"N=N1;":N1=N
    1+1
LD  1000 FORN=1TO22:VPOKEN+448,L(N+
    A):NEXT
DA  1010 PLAY"N=N1;":N1=N1+1:FORN=1
    TO21:VPOKE19+40*N,L(N+A):NEXTN,
    A
DN  1020 PLAY"S13M750T11005F4L24CCL
    16C04A05CC04A05CF3R10","S13T110
    05C4L2404AAL16AFAAFA05C3R10"
AF  1030 PLAY"L16AGFAGFAGFG12R16C24
    C4R8","L16FECFECFECE12R1604G24G
    4R8"
GC  1040 COLOR13,1,1:SCREEN1:LINE(4
    7,116)-(77,123),,BF:LINE(60,124
    )-(67,149),,BF
IO  1050 DRAW"a0s4BM47,123D5EU3RD2E
    URBM77,123D5HU3LD2HULBM41,149R7
    "
CG  1060 CIRCLE(54,149),6,,3.1416,0
    :CIRCLE(54,149),13,,3.1416,0:PA
    INT(49,160)
HN  1070 CIRCLE(94,139),8,,,,1.333:
    CIRCLE(94,139),15,,,,1.15:PAINT
    (94,150)
AF  1080 LINE(114,124)-(121,141),,B
    F:LINE(133,124)-(140,141),,BF
AN  1090 CIRCLE(127,141),6,,3.1416,
    0:CIRCLE(127,141),13,,3.1416,0:
    PAINT(127,152)
BL  1100 CIRCLE(160,137),6,,0,4.713
    :CIRCLE(160,137),13,,0,4.713:CI
    RCLE(161,157),6,,3.1416,1.571
FM  1110 CIRCLE(161,157),13,,3.1416
    ,1.571:DRAW"BM167,137R7BM147,15
    7R7":PAINT(160,169)
EF  1120 LINE(177,116)-(210,123),,B
    F:LINE(190,124)-(197,160),,BF:D
    RAW"BM177,123D5EU3RD2EURBM210,1
    23D5HU3LD2HUL"
EL  1130 COLOR15:LINE(30,179)-(221,
    181),,BF:DRAW"BM105,178L7UR3L3U

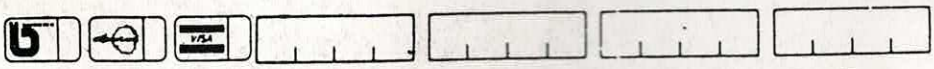
    BM105,182L7DR3L3DBM222,180R2"
MH  1140 PLAY"C4L2004AAL10A05CF4R8L
    7GFG5A16A2","A4L20FFL10FA05C4R8
    L7EDE5F16F2"
EN  1150 DRAW"C15BM62,30H2LUL3GR2L2
    GR6FL5G2D9G2LGLGL2GL2GD6RUFRFL5
    U6H2L2HL2HLHLH2ERERERERERERU7EU5ER
    2FD3G2F2D2F2GDFDF2RFRE3UEU6R2G"
    :PAINT(35,40):LINE(30,35)-(67,3
    5)
GF  1160 LINE(121,70)-(135,15):LINE
    -(150,30):LINE-(131,30):PAINT(1
    40,23)
EO  1170 COLOR10:DRAW"C10BM193,30E2
    RUR3FL2R2FL6GR5F2D9F2RFRFR2FR2F
    D6LUGLGR5U6E2R2ER2ERERE2HLHLHLH
    LHU7HU5HL2GD3F2G2D2G2FDGDG2LGLH
    3UHU6L2F":PAINT(220,40):LINE(22
    5,35)-(188,35)
HJ  1180 LINE(134,70)-(120,15):LINE
    -(105,30):LINE-(124,30):PAINT(1
    17,23)
BA  1190 DEFUSR=18312:A=USR(0):FORN
    =1TO2000:NEXTN:SCREEN1:GOTO20
AF  1200 IFPLAY(1)THEN1200ELSESCREE
    N1:GOTO20
```

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